AEMVC: Mitigate Imbalanced Embedding Space in Multi-view Clustering

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Code available at https://github.com/Lummer-Li/AEMVC

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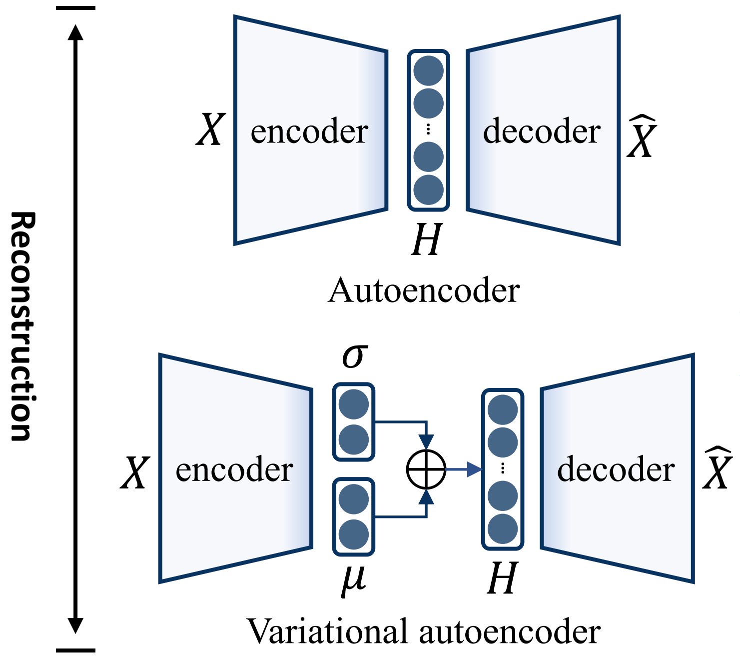
Overview



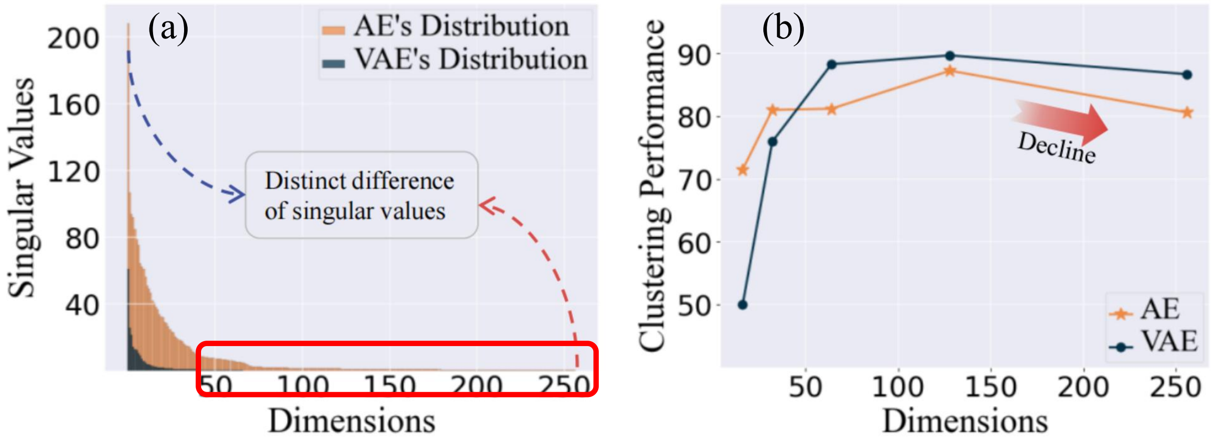


Figure 1: Illustration of key observations behind our method. These visualizations reveal fundamental bottlenecks in multi-view representation learning.

Challenge

Skewed distribution of embedding space introduces two key challenges.

➢ **Unbalanced feature contribution**.

The long-tailed distribution of singular values indicates that only a few directions contribute to the features.

➢ **Redundancy from excessive feature directions**.

Beyond a certain point, additional feature directions no longer capture meaningful information. These superfluous directions interfere with the active feature directions and degrade the embedding quality.

Overview

Motivation

Underlying mechanism of encoder-decoder-based methods obtain two cores.

- **Active feature direction**.

Data points become increasingly separable as more active feature directions are introduced.

- **Inactive feature direction**.

When inactive feature directions are active, the overlap samples are scattered.

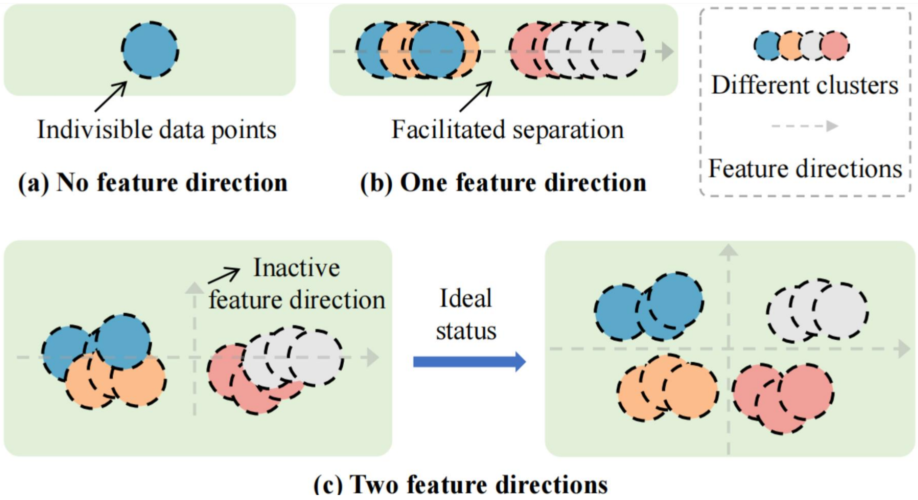


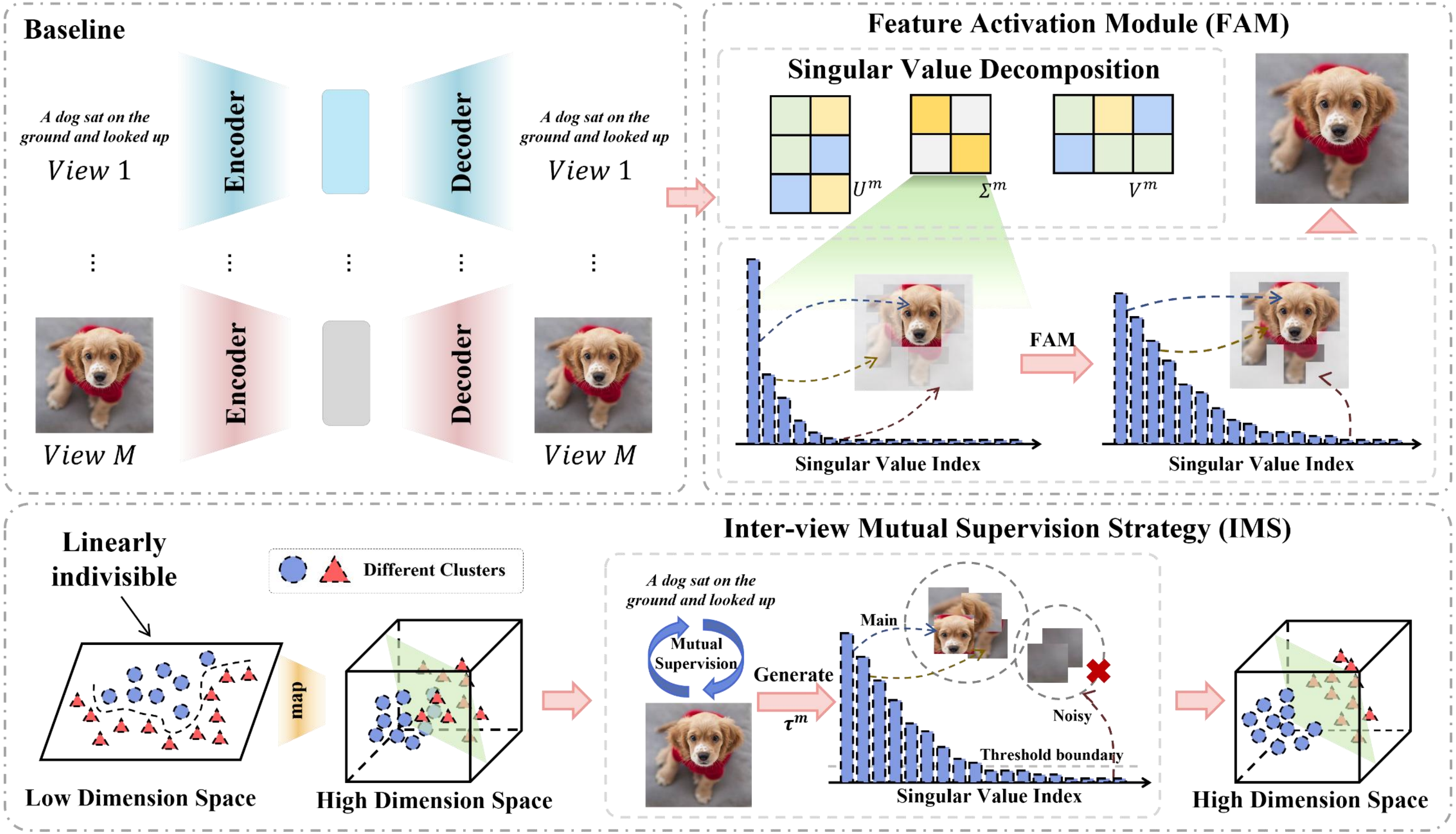
Figure 2: Conceptual illustration of feature direction contribution in clustering. This motivates us to consider multi-view clustering from the feature activation perspective.

Contribution

AEMVC: Activate-Then Eliminate Strategy for Multi-View Clustering

- We found that the embedding space learned using the encoder-decoder architecture cannot embrace the efficacy of different feature directions.
- We construct a feature activation module to adjust the contribution of different feature directions dynamically.
- We design an inter-view mutual supervision strategy to generate the view-specific thresholds, thus filtering out the redundant information.

Method



[1] Pengyuan Li, Man Liu, Dongxia Chang*, Yiming Wang, Yao Zhao. AEMVC: Mitigate Imbalanced Embedding Space in Multi-view Clustering. ACM Multimedia 2025, https://doi.org/10.1145/3746027.3754697.

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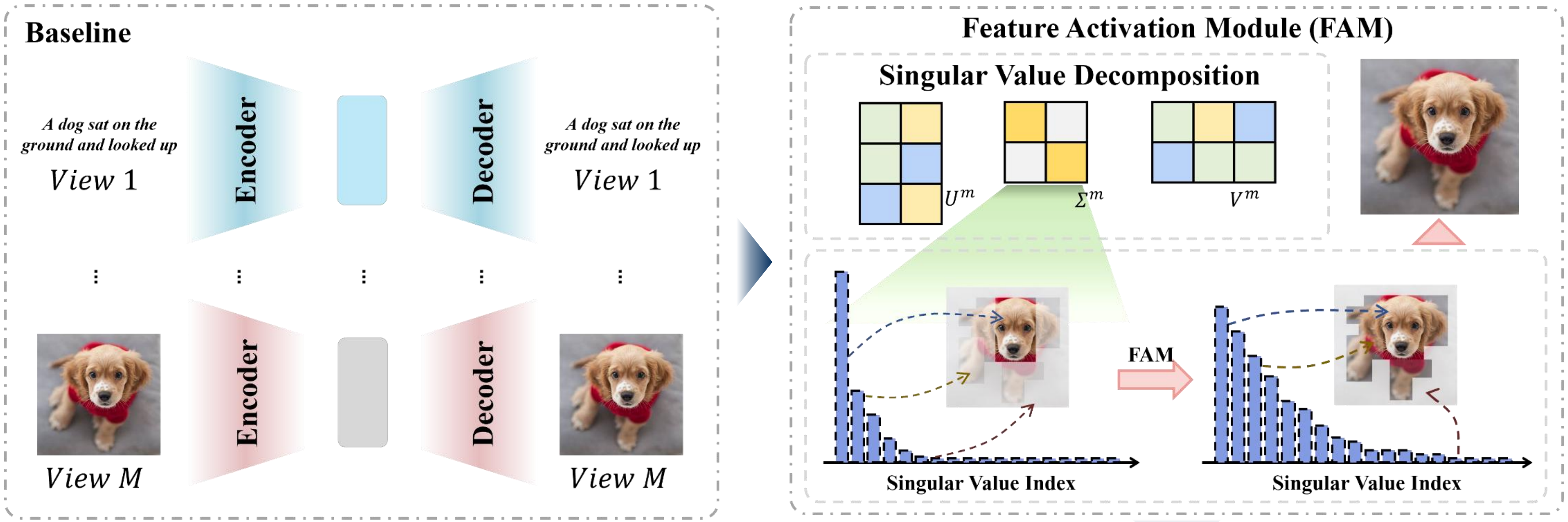

Multi-view Representation Learning

- ✓ Extract the embeddings of each view and reconstruct them.

  $\hat{X}^m = g^m(Z^m; \eta^m); Z^m = f^m(X^m; \theta^m)$
- ✓ Multi-view reconstruction loss.

$$\mathcal{L}_{rc} = \sum_{m=1}^{M} \left\| X^m - \hat{X}^m \right\|_F^2$$

Feature Activation Module

- ✓ Perform singular value decomposition for each Z_m

$$Z^m = U^m \Sigma^m (V^m)^T$$

- ✓ Encourage all non-zero singular values in Σ^m to approximate the maximum singular value σ [1]

$$\mathcal{L}_{fa} = \sum_{m=1}^{M} \sum_{i=1}^{d} (\sigma_i^m - \sigma_1^m)^2$$

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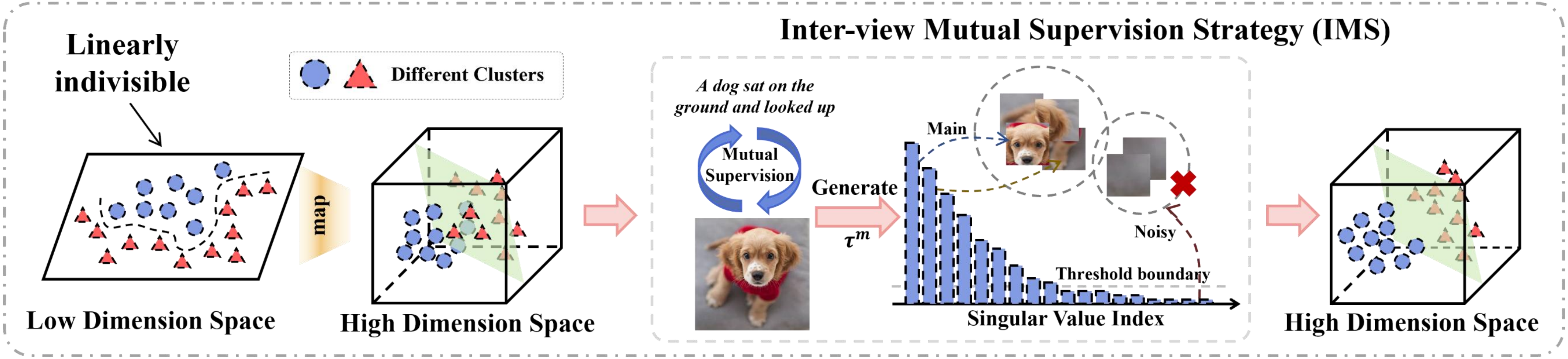
Method



Inter-view Mutual Supervision Strategy

- ✓ For view-specific representation Z^m, we perform singular value decomposition and employ a view-specific learnable threshold τ^m to generate the corresponding denoised representation.

$$\tilde{Z}^m = U^m(\Sigma^m - \tau^m I)(V^m)^T$$

- ✓ To enable adaptive threshold learning, we construct a mutual supervision objective L_{ms} to optimize each τ^m by leveraging cross-view consistency.

$$\mathcal{L}_{ms} = \sum_{m=1}^{M} \underbrace{\sum_{w=1, w\neq m}^{M} ||\tilde{Z}^m - \mathcal{P}(\tilde{Z}^w)||_F^2}_{Cross-view\ Consistency} + \underbrace{\alpha||\tilde{Z}^m||_*}_{View-specific\ Sparsity}$$

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Experimental results

	Methods	MSRCV1			Synthetic3d			UCI-Digit			ALOI			Handwritten			Scene15			Animal		
		ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI	ACC	NMI	ARI
Incomplete	DSIMVC (CVPR'22) [28]	61.81	57.57	45.30	64.97	42.86	38.40	**84.75**	78.42	72.00	58.74	72.57	39.35	77.77	78.73	69.74	25.89	26.09	12.42	17.32	17.75	5.37
	ProImp (IJCAI'23) [18]	71.33	59.83	50.34	51.00	20.68	20.06	80.54	74.21	67.68	73.01	74.41	62.37	80.54	74.21	67.68	41.44	42.69	25.00	11.82	7.41	2.48
	SCM (IJCAI'24) [24]	75.90	63.57	55.20	93.77	77.75	82.37	81.39	74.39	67.54	14.24	1.78	0.06	82.17	73.35	67.67	34.58	31.64	18.20	16.66	13.09	5.00
	RPCIC (MM'24) [49]	89.33	81.13	77.65	75.17	51.43	48.45	45.06	55.72	33.44	66.89	80.16	62.37	88.38	80.75	77.12	37.98	41.22	22.91	19.05	15.96	6.78
	DIVIDE (AAAI'24) [23]	70.67	60.07	51.23	66.03	37.13	34.20	75.42	70.84	62.09	74.33	72.78	57.34	71.15	69.40	59.17	42.99	41.54	24.92	14.61	10.05	3.14
	ICMVC (AAAI'24) [4]	74.38	64.61	56.22	84.50	55.49	59.89	83.38	**79.75**	**74.16**	82.97	79.11	71.29	83.37	79.68	74.11	41.20	43.01	24.49	11.65	6.34	1.29
	AEMVC (Ours)	**90.38**	**84.32**	**80.19**	**95.23**	**82.74**	**86.05**	83.87	78.02	62.18	**86.75**	**83.22**	**76.14**	**97.34**	**94.69**	**94.01**	**54.03**	**60.76**	**25.37**	**45.30**	**51.35**	**14.20**
Complete	DSIMVC (CVPR'22) [28]	68.95	58.72	49.67	54.50	33.49	28.30	90.23	83.09	80.02	79.35	84.92	72.12	83.58	79.44	73.51	41.73	43.31	24.83	19.23	15.55	6.74
	ProImp (IJCAI'23) [18]	84.67	79.07	72.13	90.07	69.70	73.51	82.92	81.71	75.14	75.61	80.22	65.45	82.92	81.71	75.14	41.92	43.79	25.61	13.88	10.45	3.83
	SCM (IJCAI'24) [24]	78.29	65.68	58.86	44.07	6.51	6.05	83.69	73.38	68.81	13.68	1.47	0.10	76.32	67.89	59.78	30.68	28.18	14.79	15.07	10.46	3.60
	RPCIC (MM'24) [49]	88.57	79.36	76.08	97.47	89.14	92.60	89.03	81.71	77.80	50.96	66.33	35.87	83.36	73.85	67.52	27.86	27.23	13.49	19.58	18.14	6.86
	DIVIDE (AAAI'24) [23]	91.52	85.31	82.26	90.67	70.18	74.63	90.10	86.06	82.42	88.06	85.16	72.33	96.92	93.14	93.29	49.03	49.31	31.93	16.49	12.67	4.16
	ICMVC (AAAI'24) [4]	93.90	88.69	86.78	84.70	57.56	61.46	83.00	81.97	74.81	89.99	86.30	81.73	83.41	82.38	75.45	41.26	43.92	25.57	13.07	10.80	3.40
	VITAL (MM'24) [12]	91.90	86.34	83.53	97.53	89.06	92.77	89.43	86.36	82.08	96.26	93.43	91.91	84.18	82.57	76.06	43.85	46.44	28.37	17.14	13.48	5.99
	SCMVC (TMM'24) [41]	87.43	79.35	74.80	96.37	85.16	89.49	71.65	69.32	59.39	15.24	2.29	0.30	82.53	73.77	66.68	39.13	39.16	24.51	18.50	14.95	6.38
	DCMVC (TIP'24) [6]	82.00	72.99	66.09	**98.50**	**92.93**	**95.56**	94.57	90.15	88.53	96.53	**97.00**	95.35	88.57	90.77	86.31	45.36	44.83	29.11	22.97	21.32	10.95
	AEMVC (Ours)	**95.33**	**91.36**	**89.46**	97.53	89.58	92.71	**94.70**	**91.05**	**88.62**	**97.96**	95.69	**95.46**	**98.67**	**96.73**	**97.06**	**58.72**	**67.21**	**37.58**	**50.27**	**56.53**	**20.02**

Table 1: Clustering results on seven multi-view datasets. "Incomplete" represents data with 50% missing views, while "Complete" denotes data without missing views. The best and second-best results are highlighted in bold and underlined, respectively. Note that VITAL, SCMVC, and DCMVC are evaluated only on complete data as they are not designed for incomplete scenarios.

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Experimental results

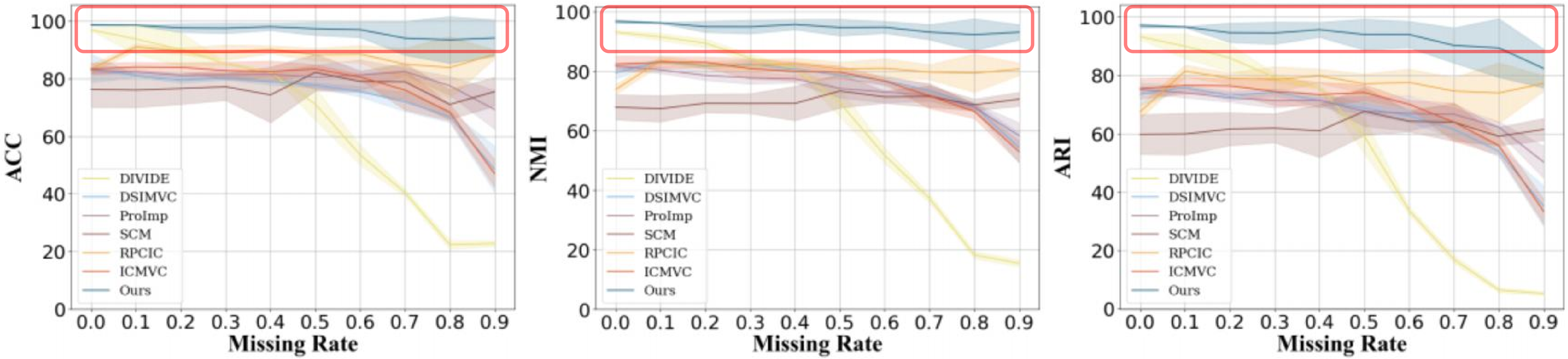

Figure 3: Clustering performance under different missing rates on Handwritten dataset. The colored regions denote the standard variances with five random experiments.

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Experimental results

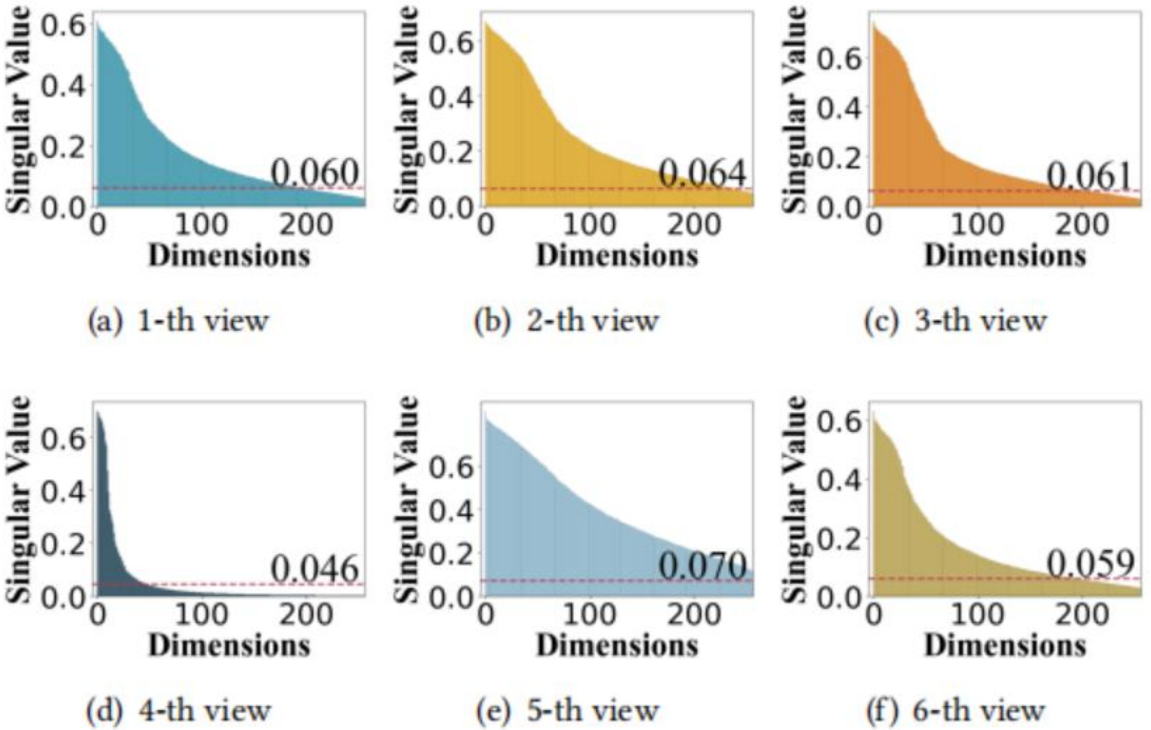

Figure 4: The singular value distribution for each view on the Handwritten dataset after embeddings are activated. The dashed lines indicate the adjusted thresholds learned via IMS.

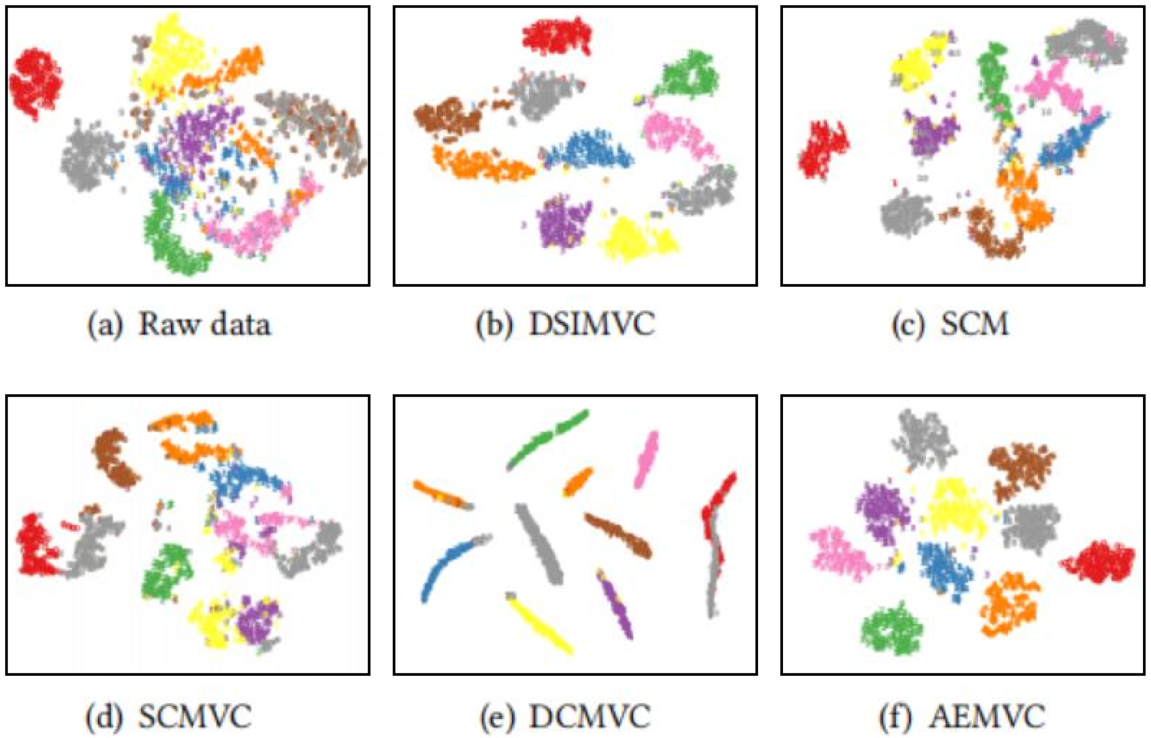

Figure 5: Comparison of t-SNE visualizations learned by different methods on the UCI-Digits dataset.

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Thanks !

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